INCH-POUND

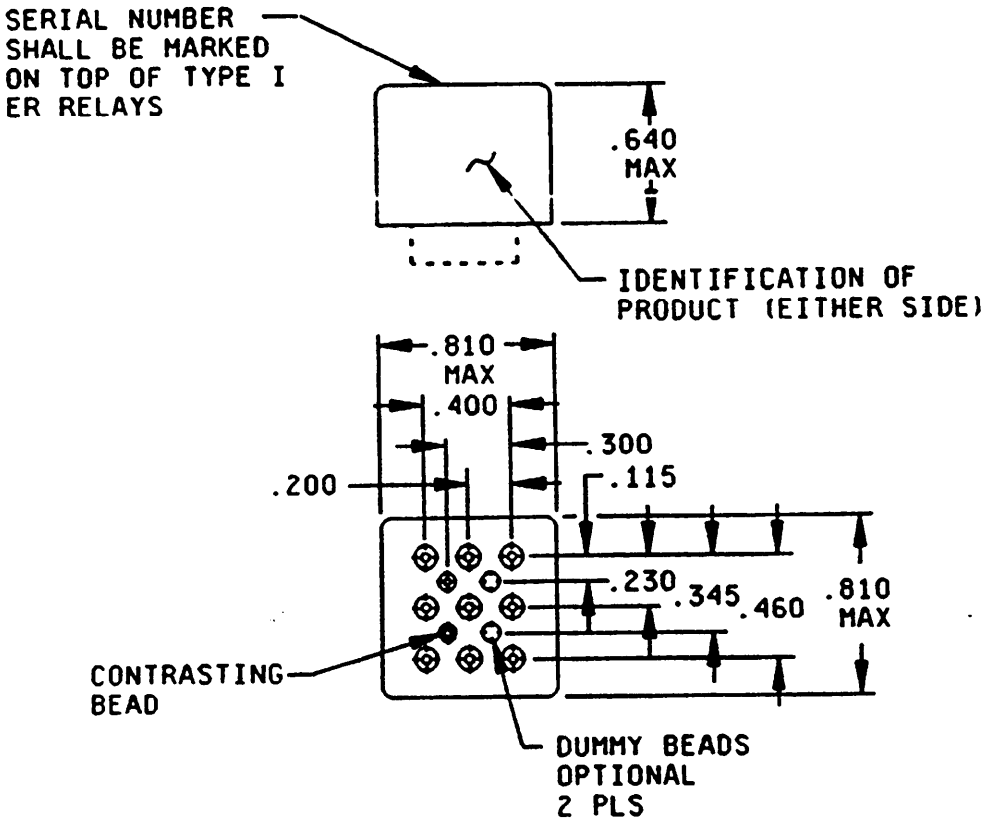
MIL-R-6106/29E
1 July 1993
SUPERSEDING
MIL-R-6106/29D
21 December 1990

MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, TYPE I, (INCLUDING ESTABLISHED RELIABILITY (ER) TYPES), PERMANENT MAGNET DRIVE, LOW LEVEL TO 10 AMPERES, 3 PDT, HERMETICALLY SEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-6106.

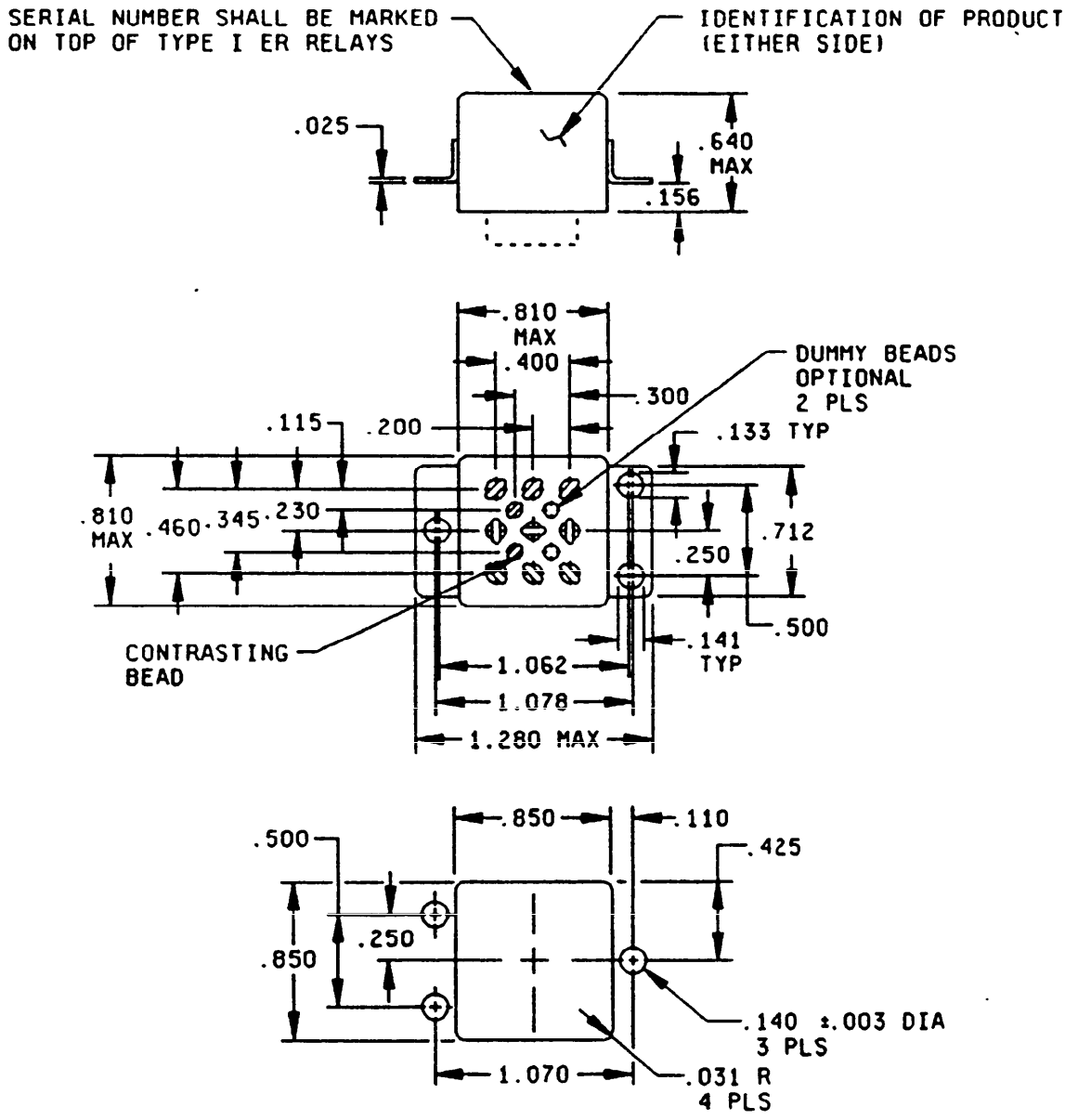


CONFIGURATION A

FIGURE 1. Configurations, dimensions, and circuit diagrams.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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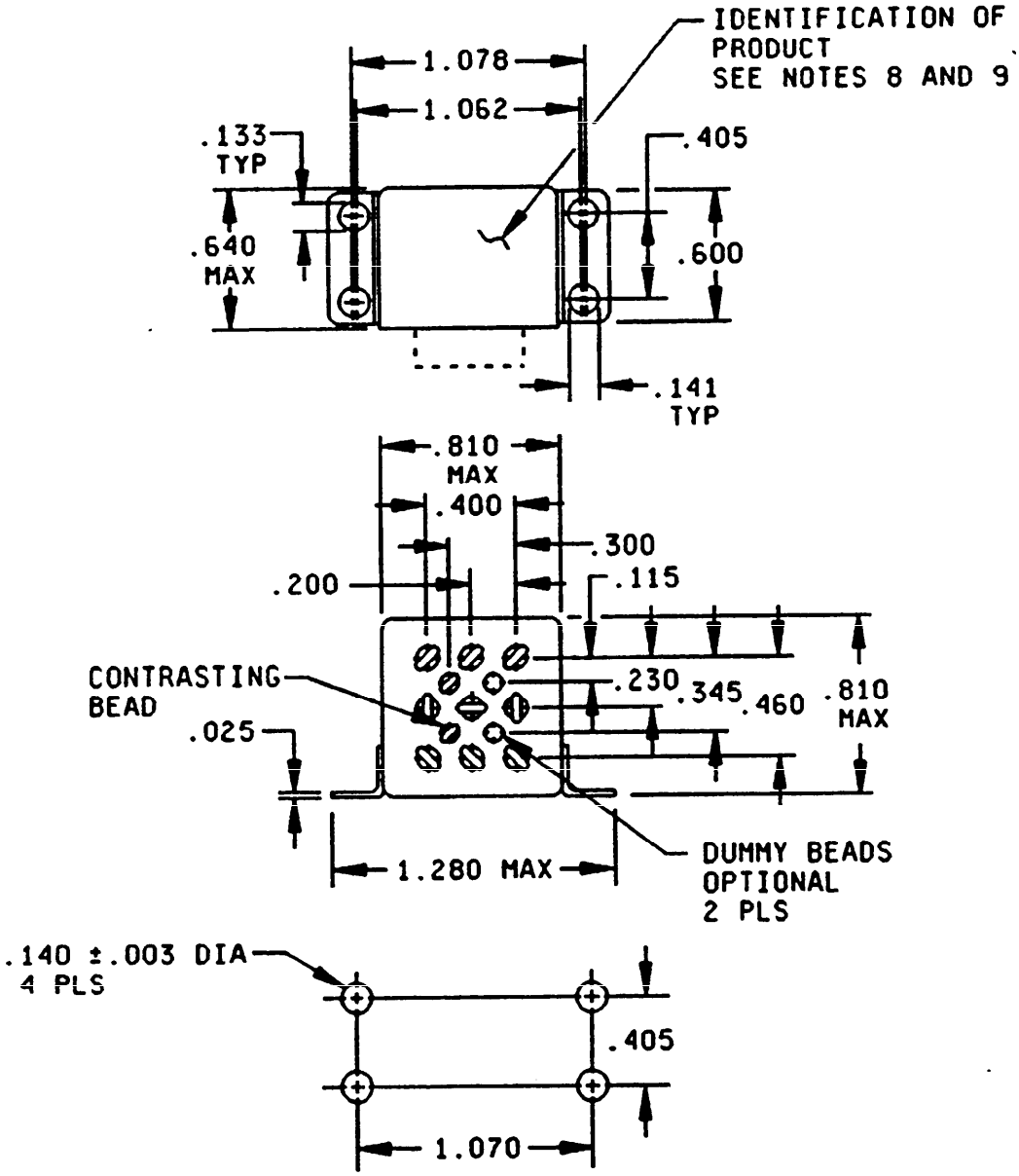


MOUNTING HOLE LAYOUT

CONFIGURATION B

FIGURE 1. Configurations, dimensions, and circuit diagrams - Continued.

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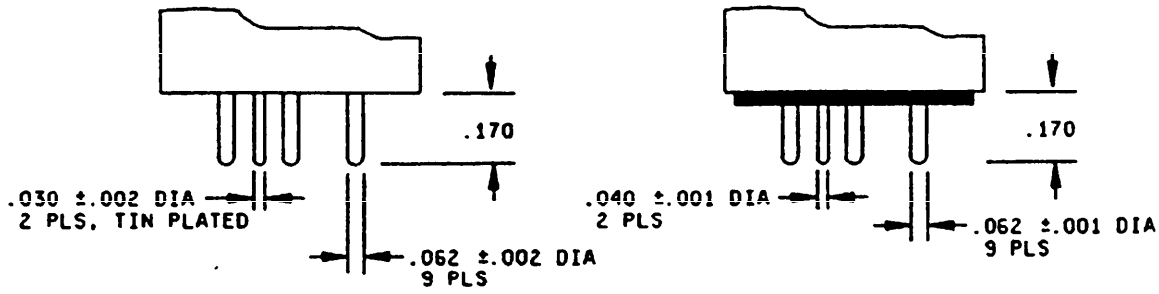


MOUNTING HOLE LAYOUT

CONFIGURATION C

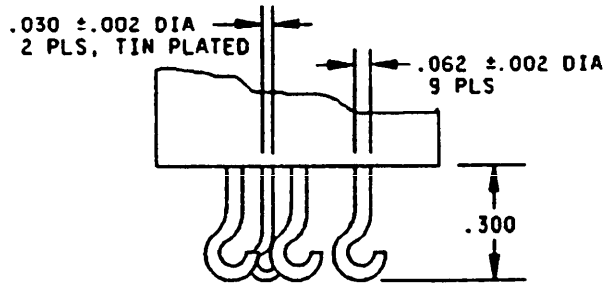
FIGURE 1. Configurations, dimensions, and circuit diagrams - Continued.

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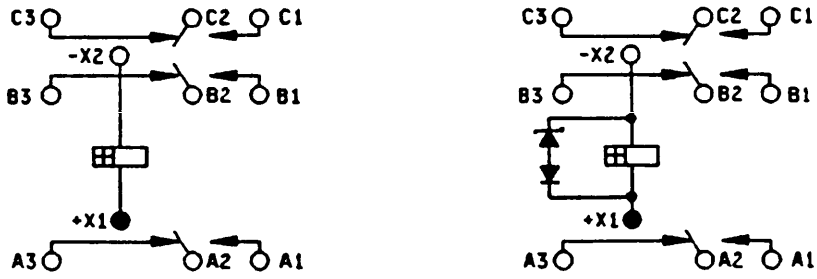
SOLDER PIN

SOCKET PIN
SEE NOTES 10 AND 11



SOLDER HOOK

TERMINALS



TYPE A

TYPE B

CIRCUIT DIAGRAM

FIGURE 1. Configurations, dimensions, and circuit diagrams - Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	0.03	.110	2.79	.230	5.84	.600	15.24
.002	0.05	.115	2.92	.250	6.35	.640	16.26
.003	0.08	.133	3.38	.300	7.62	.712	18.08
.025	0.64	.140	3.56	.345	8.76	.810	20.57
.030	0.76	.141	3.58	.400	10.16	.850	21.59
.031	0.79	.156	3.96	.405	10.29	1.062	26.97
.040	1.02	.170	4.32	.425	10.80	1.078	27.38
.050	1.27	.200	5.08	.500	12.70	1.280	32.51

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 inch (0.25 mm).
4. There shall be affixed to the relay a legible circuit diagram that identifies each terminal location specified. These relays are polarized monostable.
5. This relay shall not operate or be damaged by reverse polarity. Semiconductors shall not be used for this purpose.
6. Permanent magnet drive consists of a permanent magnet with its flux path switched and combined with the electromagnetic flux.
7. Applicable to configuration C only. The circuit diagram, manufacturer's PIN, and the military PIN shall be marked on the near side. The remaining portion of the nameplate data shall be marked on the far side.
8. Applicable to configuration C only. Relays shall be marked with the manufacturer's name or source code and data code. In addition, ER relays shall be marked with the serial number. Marking shall be with the bottom of print adjacent to the near side.
9. Socket pin terminals shall be gold-plated in accordance with MIL-G-45204, type II, class I. The nickel underplating shall be 50 to 150 microinches thick (minimum).
10. Silicone rubber gasket, AMS 3332, shore hardness 25 ±5, thickness .050 ±.005 inch (1.27 ±0.13 mm).
11. Back EMF (transient voltage): 42 V dc maximum.
12. JANTX or equivalent screened semiconductor shall be used in ER relays with internal coil suppression. Relays using suppression devices shall continue to operate should the suppression circuit be in a failure mode. Diodes shall have a peak inverse voltage of 600 V dc minimum.

FIGURE 1. Configurations, dimensions, and circuit diagrams - Continued.

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REQUIREMENTS:

CONFIGURATIONS, DIMENSIONS, AND CIRCUIT DIAGRAMS: See figure 1.

CONTACT DATA:

Life: Unless otherwise specified: 50,000 cycles.

Load ratings:

High level (relay case grounded):

Resistive: 10 amperes at 28 V dc; 115 V ac, 400 Hz, 1 phase; and 115/200 V ac, 400 Hz, 3 phase.

Inductive: 6 amperes at 28 V dc (life: 10,000 cycles); 8 amperes at 115 V ac, 400 Hz, 1 phase and 115/200 V ac, 400 Hz, 3 phase. (Life: 20,000 cycles.)

Motor: 4 amperes at 28 V dc; 115 V ac, 400 Hz, 1 phase; and 115/200 V ac, 400 Hz, 3 phase.

Lamp: 2 amperes at 28 V dc and 115 V ac, 400 Hz, 1 phase.

Low level: 10 to 50 μA at 10 to 50 mV or peak ac (life: 100,000 cycles). 1/ 2/ 3/

Intermediate current: Applicable.

Transfer load: Not applicable.

Mechanical life (reduced current): 1.25 amperes at 28 V dc, 115 V ac, 1 phase and 3 phase (life: 200,000 cycles.)

Mixed loads: Applicable. 4/

Contact voltage drop or resistance: 1/ 2/ 3/

High level:

Initial: 0.150 volt maximum.
After life: 0.175 volt maximum.

Low level:

Initial: 0.050 ohm maximum.
After life: 0.150 ohm maximum.

Contact bounce: 1.0 millisecond maximum.

Break bounce (normally open contacts only): 0.1 millisecond maximum (applicable to dc suppressed coils only).

1/ For low level, the following shall apply:
   a. During endurance and operational reliability testing, the contact load shall be 10 to 50 microamperes, 10 to 50 millivolts open circuit voltage, and 100 ohms maximum contact resistance.
   b. Static contact resistance shall be performed at 50 mA maximum and 50 mV maximum.
2/ For all ER relays, the alternate low level test of operational reliability shall be used.
3/ For group A contact voltage drop test, high level testing shall be performed first, followed by low level testing. The contacts shall not make or break the high level load.
4/ Relay shall be capable of switching low level while switching any of the other rated loads on adjacent contacts.

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Overload current: 30 amperes dc.
60 amperes ac.

Rupture current: 40 amperes dc.
80 amperes ac.

Time current relay characteristics: Applicable (see table I).

TABLE I. Time current relay characteristics. 1/ 2/

1	20 A	1 hour
2	30 A	20 seconds
3	75 A	2 seconds
4	150 A	.53 second

1/ CAUTION: Compare with time current characteristics of the associated circuit protective devices.

2/ All relays shall withstand overload and fault currents. Relays must be able to sustain five applications (make and carry only) of power concurrently on adjacent poles at each of four different current levels in the sequence listed in table I. Separate relays shall be tested at 28 V dc and 115/200 V ac, 400 Hz, 3 phase. Cooling time between successive applications shall be 30 minutes. Tests shall be performed on both normally open and normally closed contacts. There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at the conclusion.

COIL DATA 5/: See table II.

Operate time: 6 milliseconds maximum with rated coil voltage.

Release time: 6 milliseconds maximum from rated coil voltage.

Duty rating: Continuous.

5/ CAUTION: Due to possible interaction of relay magnetic fields, the following spacing requirements, as a minimum, shall be considered in dense packaging situations:
a. Row to row assisting fields: .125 inch.
b. Row to row opposing fields: .1875 inch.
c. Side to side alternating fields: .0625 inch.
d. Side to side like fields: .125 inch.

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TABLE II. Dash numbers and characteristics. 1/

Military PIN 2/ M6106/29-	Type	Coil suppression (transient voltage, back EMF) 2/	Terminal type	Mating socket	Mounting configuration	Circuit type	Coil data								
							Rated voltage (V dc) 4/	Resistance Ω ±10% at 25°C	Maximum		Pickup voltage (maximum) (V dc)			Hold voltage (V dc) 6/	Drop-out voltage (V dc) 6/
									Voltage (V dc) 5/	Current (A)	Normal 6/	High temp test	Continuing current test		
001 3/	I	N/A	Solder pin	N/A	A	A	28	400	29	.081	18	19.8	22.5	7.0	1.5
002	I ER	N/A	"	"	"	A	"	"	"	"	"	"	"	"	"
003 3/	I	42 V dc max	"	"	"	B	"	"	"	"	"	"	"	"	"
004	I ER	42 V dc max	"	"	"	B	"	"	"	"	"	"	"	"	"
005 3/	I	N/A	Solder hook	"	"	A	"	"	"	"	"	"	"	"	"
006	I ER	N/A	"	"	"	A	"	"	"	"	"	"	"	"	"
007 3/	I	42 V dc max	"	"	"	B	"	"	"	"	"	"	"	"	"
008	I ER	42 V dc max	"	"	"	B	"	"	"	"	"	"	"	"	"
009 3/	I	N/A	Solder pin		"	A	"	"	"	"	"	"	"	"	"
010	I ER	N/A	"	MIL-S-12883/	"	A	"	"	"	"	"	"	"	"	"
011 3/	I	42 V dc max	"	46-01	"	B	"	"	"	"	"	"	"	"	"
012	I ER	42 V dc max	"		"	B	"	"	"	"	"	"	"	"	"
013 3/	I	N/A	Solder pin	N/A	B	A	"	"	"	"	"	"	"	"	"
014	I ER	N/A	"	"	"	A	"	"	"	"	"	"	"	"	"
015 3/	I	42 V dc max	"	"	"	B	"	"	"	"	"	"	"	"	"
016	I ER	42 V dc max	"	"	"	B	"	"	"	"	"	"	"	"	"
017 3/	I	N/A	Solder hook	"	"	A	"	"	"	"	"	"	"	"	"
018	I ER	N/A	"	"	"	A	"	"	"	"	"	"	"	"	"
019 3/	I	42 V dc max	"	"	"	B	"	"	"	"	"	"	"	"	"
020	I ER	42 V dc max	"	"	"	B	"	"	"	"	"	"	"	"	"
021 3/	I	N/A	Solder pin		"	A	"	"	"	"	"	"	"	"	"
022	I ER	N/A	"	MIL-S-12883/	"	A	"	"	"	"	"	"	"	"	"
023 3/	I	42 V dc max	"	46-01	"	B	"	"	"	"	"	"	"	"	"
024	I ER	42 V dc max	"		"	B	"	"	"	"	"	"	"	"	"
025 3/	I	N/A	Solder pin	N/A	C	A	"	"	"	"	"	"	"	"	"
026	I ER	N/A	"	"	"	A	"	"	"	"	"	"	"	"	"
027 3/	I	42 V dc max	"	"	"	B	"	"	"	"	"	"	"	"	"
028	I ER	42 V dc max	"	"	"	B	"	"	"	"	"	"	"	"	"
029 3/	I	N/A	Solder hook	"	"	A	"	"	"	"	"	"	"	"	"
030	I ER	N/A	"	"	"	A	"	"	"	"	"	"	"	"	"
031 3/	I	42 V dc max	"	"	"	B	"	"	"	"	"	"	"	"	"
032	I ER	42 V dc max	"	"	"	B	"	"	"	"	"	"	"	"	"

1/ For type I ER relays, a suffix letter (M, U, X, or P) to indicate the applicable failure rate level shall be added to the dash number. Failure rate levels (percent per 10,000 cycles): M = 1%, X = 0.3% and P = 0.1%.
Examples: -001M; -032P.

2/ JANTX or equivalent screened semiconductors shall be used in ER relays with internal coil suppression. Relays using suppression devices shall continue to operate should the suppression circuit be in a failure mode. Diodes shall have a peak inverse voltage of 600 V dc minimum.

3/ Type I relays are inactive for new design after 21 December 1990. See table IV for cross reference data.

4/ CAUTION: The use of any coil voltage less than the rated voltage will compromise the operation of the relay.

5/ When maximum ambient temperature does not exceed +85°C, the maximum coil voltage shall be 32 V dc.

6/ Over (temperature range.

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ELECTRICAL DATA:

Insulation resistance, initial: 100 megohms.

After life or environmental tests: 50 megohms.

Dielectric withstanding voltage (sea level):	Initial V rms (60 Hz)	After life tests V rms (60 Hz)
Coil to case	1,000	750
All other points	1,250	1,000

Dielectric withstanding voltage (altitude) 6/:	80,000 ft V rms (60 Hz)	300,000 ft V rms (60 Hz)
Coil to case	250	500
All other points	250	500

ENVIRONMENTAL CHARACTERISTICS:

Temperature range: -70°C to +125°C.

Maximum altitude rating: 300,000 feet.

Acceleration: Applicable (15 g's maximum).

Shock (specified pulse): MIL-STD-202, method 213, test condition C (100 g's), except that for configurations A and B peak g value shall be 200 g's. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum for open contacts.

Vibration (sinusoidal): Configurations A and B shall be 30 g's, 70 to 3,000 Hz. Configuration C shall be 20 g's, 57 to 3,000 Hz. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum for open contacts.

Vibration (random): MIL-STD-202, method 214 for configurations A and B, test condition IG (0.4 g^2/Hz); for configuration C, test condition IE (0.2 g^2/Hz). Duration shall be 15 minutes in each plane for all configurations. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum for open contacts.

PHYSICAL DATA:

Terminal strength:

Solder hook terminals:

Bend test: Not applicable.

Twist test: Not applicable.

Pull force (0.030 diameter): 3 ±0.3 pounds.

Pull force (0.062 diameter): 10 ±1 pounds.

6/ Dielectric may be improved by suitable insulation of terminals and wiring after installation.

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Solder pin terminals:

Bend test: Applicable.

Pull force (0.030 diameter): 3 pounds ±0.3 pounds.

Pull force (0.062 diameter): 10 pounds ±1 pounds.

Socket pin terminals:

Bend test: Applicable.

Pull force (0.040 diameter): 5 pounds ±0.5 pounds.

Pull force (0.062 diameter): 10 pounds ±1 pounds.

Terminal solderability: Applicable to solder pin and solder hook terminals only.

Weight (maximum): .062 pound (28.15 grams).

Seal: Hermetic; relays shall be sealed by welding (laser, TIG (tungsten inert gas) or other suitable means as approved by the qualifying activity).

Construction (internal and external): All welded, except that coil magnet wire to coil lead wire shall be soldered.

Identification of product: Applicable. The circuit diagram shall be oriented to that when the relay is held with the circuit diagram right side up as shown on figure 1, and rotated away from the viewer about a horizontal axis through the diagram until the header terminals face the viewer, then each terminal shall be in the location shown on the circuit diagram.

Supersession data: See table III.

TABLE III. Supersession data.

Superseded PIN M6106/29-	New PIN M6106/29-
-033	-025
-034	-026
-035	-027
-036	-028

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Cross-reference for Government logistical support: See table IV.

TABLE IV. Cross-reference for Government logistical support.

PIN M6106/29-	Support with PIN M6106/29- 1/	PIN M6106/29-	Support with PIN M6106/29- 1/
001	002	017	018
002	002	018	018
003	004	019	020
004	004	020	020
005	006	021	022
006	006	022	022
007	008	023	024
008	008	024	024
009	010	025	026
010	010	026	026
011	012	027	028
012	012	028	028
013	014	029	030
014	014	030	030
015	016	031	032
016	016	032	032

1/ A suffix letter (M, U, X, or P) to designate the applicable failure rate level shall be added to the dash number. Failure rate level (percent per 10,000 cycles): M, 1.0; U, 0.5; X, 0.3; P, 0.1. (Examples: -002M; -004P).

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Qualification by similarity: See table V.

TABLE V. Qualification by similarity. 1/

Military PIN M6106/29-	Loads		Dynamics			Environmental
	Basic qual.	ER	Sockets/Pins			
	A	D	A	B	C	A
	DC 2/ 3/	DC 2/ 3/	Configuration A 2/ 3/	Configuration B 2/ 3/	Configuration C 2/ 3/	DC 2/ 4/
001 5/	4	3	2			4
002	4	3	2			4
003 5/	1, 2, 3	1, 2	1			1, 2, 3
004	2, 3	1, 2	1			2, 3
005 5/	4	3	2			4
006	4	3	2			4
007 5/	1, 2, 3	1, 2	1			1, 2, 3
008	2, 3	1, 2	1			2, 3
009 5/	4	3	2			4
010	4	3	2			4
011 5/	1, 2, 3	1, 2	1			1, 2, 3
012	2, 3	1, 2	1			2, 3
013 5/	4	3		2		4
014	4	3		2		4
015 5/	1, 2, 3	1, 2		1		1, 2, 3
016	2, 3	1, 2		1		2, 3
017 5/	4	3		2		4
018	4	3		2		4
019 5/	1, 2, 3	1, 2		1		1, 2, 3
020	2, 3	1, 2		1		2, 3
021 5/	4	3		2		4
022	4	3		2		4
023 5/	1, 2, 3	1, 2		1		1, 2, 3
024	2, 3	1, 2		1		2, 3
025 5/	4	3			2	4
026	4	3			2	4
027 5/	1, 2, 3	1, 2			1	1, 2, 3
028	2, 3	1, 2			1	2, 3
029 5/	4	3			2	4
030	4	3			2	4
031 5/	1, 2, 3	1, 2			1	1, 2, 3
032	2, 3	1, 2			1	2, 3

1/ See amendment 1, MIL-R-6106, for guidelines for determining how relays are to be grouped and ranked within each subgroup.

2/ DC suppressed coils: Discretes qualify only discretes; hybrids qualify only hybrids.

3/ Terminals: Socket pin type terminals are considered worst case. Solder pins and hooks may be qualified by similarity.

4/ Socket pins (with gasket seals) are considered nonsimilar to solder pins and hooks and must be tested separately.

5/ Inactive for new design after 21 December 1990.

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Quality conformance inspection:

Performance of group C tests is not applicable to type I relays.

Marginal notations are not used in this revision to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-0890-03)

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